Week 7

Deep Learning (IIT Ropar)

Data for questions from 1 to 5

Air Quality Index (AQI) Prediction for an Indian Smart City Using Ensemble Models

Air pollution has become a major public health concern in several Indian cities, particularly during winter months. As part of the Smart Cities Mission, the municipal corporation of Delhi initiates a data-driven project to forecast the daily Air Quality Index (AQI) so that early warnings can be issued to vulnerable populations such as children, elderly citizens, and patients with respiratory conditions.

The dataset for this project is collected from continuous monitoring stations deployed across the city over a period of five years. Each data record corresponds to one day and includes the following features:

- Concentration of PM2.5 (µg/m³)
- Concentration of PM10 (µg/m³)
- Nitrogen dioxide (NO_2) levels
- Sulphur dioxide (SO_2) levels
- Wind speed
- Ambient temperature

The target variable is the daily AQI value, treated as a continuous regression output.

To build a reliable prediction system, the data science team experiments with three different predictive models, each trained on the same training dataset of 10,000 samples and evaluated on a test dataset of 2,000 unseen samples:

- **Model A:** Linear Regression
- **Model B:** Random Forest Regressor
- **Model C:** Feedforward Neural Network

After training, the following test-set Mean Squared Errors (MSE) are observed:

Model	**Test MSE**
Model A (Linear Regression)	210
Model B (Random Forest)	165
Model C (Neural Network)	180

Although Model B performs better than the others individually, the team notices that the prediction errors made by the three models are not identical. For some days, the linear model underestimates AQI during pollution spikes, while the neural network sometimes overestimates AQI on cleaner days. The random forest performs well on average but is sensitive to certain outlier conditions. To exploit this diversity, the team constructs an ensemble model by averaging the predictions of all three models:

$$\hat{y}_{ensemble} = \frac{1}{3}(\hat{y}A + \hat{y}B + \hat{y}C)$$

After deploying this ensemble on the same test dataset, the observed ensemble MSE reduces to 140, which is significantly lower than that of any individual model. To further analyze prediction stability, the team evaluates the models on four different random subsets of the training data and records the predicted AQI for a particular day with severe pollution. The predictions are:

- Model A predictions: [310, 295, 305, 300]
- Model B predictions: [330, 340, 325, 335]
- Model C predictions: [290, 285, 300, 295]

The true AQI value on that day is **320**.

The municipal authorities must now decide whether the ensemble model is more suitable for deployment, especially given the need for robustness, reduced variance, and reliable generalization under varying environmental conditions.

Question: 1

What is the primary advantage of using an ensemble model for AQI prediction in this case?

Options:

A) It guarantees zero prediction error
B) It reduces model bias
C) It reduces variance by combining diverse models
D) It eliminates the need for training data

Correct Answer: C

Explanation: Ensembles reduce variance by averaging out errors from different models that make diverse mistakes.

Question: 2

Despite Model B having the lowest individual MSE, why does the ensemble outperform it?

Options:

A) Model B is underfitting
B) Ensemble combines uncorrelated errors
C) Ensemble increases model complexity
D) Model B has the highest bias

Correct Answer: B

Explanation: When individual models make different errors, averaging them reduces overall error.

Question: 3

What is the average prediction of the ensemble for the severe pollution day, using the mean prediction of each model?

A) 300
B) 305
C) 309
D) 320

Correct Answer: C

Explanation:

- Model A mean = (310 + 295 + 305 + 300)/4 = 302.5
- Model B mean = (330 + 340 + 325 + 335)/4 = 332.5
- Model C mean = (290 + 285 + 300 + 295)/4 = 292.5
- $\hat{y}_{ensemble} \approx 309$

Question: 4

Which of the following are **valid reasons** for deploying the ensemble AQI model in a Smart City context?

Options:

A) Improved robustness to unusual pollution patterns
B) Reduced sensitivity to noisy sensor readings
C) Guaranteed perfect AQI forecasts
D) Better generalization across seasons

Correct Answers: A, B, D

Explanation: Ensembles improve robustness and generalization but cannot guarantee perfect predictions.

Question: 5

Compared to individual models, the ensemble model is most likely to have:

Options:

A) Lower variance
B) Higher variance
C) Similar or slightly higher bias
D) Improved generalization

Correct Answers: A, C, D

Explanation: Ensembles mainly reduce variance, sometimes at the cost of a small bias increase.

Data for questions from 6 to 10

Crop Disease Detection Using Dataset Augmentation in Indian Agriculture

Agriculture plays a vital role in India's economy, and crop diseases are a major cause of yield loss for farmers. In Maharashtra, an AgriTech startup collaborates with the state agriculture department to build a deep learning–based crop disease detection system that can identify diseases from leaf images captured using mobile phones by farmers.

Due to practical constraints, the startup is initially able to collect only 1,200 labeled images of crop leaves across four disease categories and one healthy category. These images are captured under varying field conditions such as different lighting, angles, and background clutter. Because the dataset is relatively small, the data science team is concerned about overfitting when training a convolutional neural network (CNN).

A CNN model is trained using 70% of the data for training and 30% for validation. After training for several epochs without any special technique, the team observes the following:

- Training accuracy reaches 98%
- Validation accuracy remains around 72%
- Validation loss fluctuates significantly

To improve the model's generalization ability, the team decides to apply dataset augmentation techniques during training. The augmentations are applied independently to each training image: They are Horizontal flip, Random rotation within ±15°, Random zoom (up to 10%), Each augmented version is treated as a new training sample.

After applying augmentation, the model is retrained using the same architecture and hyperparameters. This time, the team observes that the **training accuracy slightly decreases**, but the **validation accuracy improves significantly**, and the validation loss becomes more stable.

The team must now evaluate whether augmentation has successfully reduced overfitting, increased robustness to real-world variations, and improved the model's suitability for deployment in real agricultural settings where images are captured under uncontrolled conditions.

Question: 06

The large gap between training accuracy (98%) and validation accuracy (72%) before augmentation indicates:

Options:

A) High bias
B) High variance
C) Data leakage
D) Underfitting

Correct Answer: B

Explanation: High training accuracy with low validation accuracy is a classic symptom of overfitting, which corresponds to high variance.

Question: 07

Out of the original 1,200 images, how many images are used for training before augmentation?

Options:

A) 720
B) 840
C) 900
D) 1,200

Correct Answer: B

Explanation: Training data = 70% of 1,200 = **840 images**.

Question: 08

If each training image generates three additional augmented versions, what is the effective size of the training dataset after augmentation?

Options:

A) 1,680
B) 2,520

C) 3,360
D) 4,200

Correct Answer: C

Explanation:
Each original image produces 1 original + 3 augmented = 4 images
Effective training size = 840 × 4 = **3,360**

Question: 09

Why does training accuracy slightly decrease after applying dataset augmentation?

Options:

A) The model capacity is reduced
B) Augmentation introduces label noise
C) The learning rate is too small
D) The task becomes harder due to increased variability

Correct Answer: D

Explanation: Augmentation introduces realistic variations, making training harder and preventing memorization.

Question: 10

Which augmentation strategies used in this case are appropriate for crop leaf images?

Options:

A) Horizontal flipping
B) Rotation within ±15°
C) Vertical flipping
D) Random zoom

Correct Answers: A, B, D

Explanation: Small rotations and zoom simulate realistic camera variations. Vertical flipping may create unrealistic leaf orientations in some crops.

Data for questions from 11 to 15

Credit Card Fraud Detection Using L2 Regularization in an Indian FinTech System

With the rapid growth of digital payments in India, especially through credit cards and mobile banking platforms, financial fraud has become a serious concern for banks and FinTech companies. A Bengaluru-based FinTech startup is developing a machine learning–based fraud detection system to identify suspicious credit card transactions in real time.

The dataset used for this task is collected from transaction logs of multiple Indian banks and contains 50,000 historical transactions, each labeled as either legitimate (0) or fraudulent (1). Fraudulent transactions form only 3% of the dataset, making the problem highly imbalanced. Each transaction record includes numerical features such as Transaction amount, Time since last transaction, Merchant risk score, Number of transactions in the past 24 hours, and Device risk score

To model the probability of fraud, the data science team initially uses a logistic regression model with multiple input features. The model is trained using gradient descent on 80% of the dataset, while the remaining 20% is reserved for testing. After training the initial model without any regularization, the team observed a training accuracy of 99% and a test accuracy of 86%.

Although the training accuracy is very high, the relatively lower test accuracy indicates that the model may be overfitting the training data. Further inspection reveals that the learned weights have very large magnitudes, especially for features related to transaction amount and merchant risk.

To address this issue, the team introduces L2 regularization (Ridge) into the training objective. The new loss function becomes:

$$J(w) = Loss(w) + \alpha \sum_i w_i^2$$

where α is the regularization strength that controls the penalty on large weights. After retraining the model with $\boldsymbol{\alpha = 2}$, the following changes are observed:

- Training accuracy decreases slightly to **95%**
- Test accuracy improves to **92%**
- Learned weights become significantly smaller in magnitude

For one particular feature, the model parameters before and after regularization are:

- **Without L2:** $w = 6.0$
- **With L2:** $w = 3.5$

During one iteration of training with L2 regularization, the following values are used:

- Gradient of loss (without regularization): $\nabla J = 4$
- Learning rate: $\eta = 0.1$
- Current weight value: w=5
- Regularization strength: α=2

The FinTech company must now evaluate how L2 regularization has affected model complexity, bias–variance tradeoff, and generalization, and whether this model is suitable for deployment in a real-world financial system where incorrect predictions can have serious consequences.

Question: 11

The initial model (without regularization) shows very high training accuracy but lower test accuracy. This behavior most strongly indicates:

Options:

A) Underfitting
B) High bias
C) Overfitting
D) Data leakage

Correct Answer: C

Explanation: A large gap between training and test accuracy indicates that the model has memorized training data and fails to generalize.

Question: 12

What is the primary role of L2 regularization in this fraud detection model?

Options:

A) It removes irrelevant features
B) It shrinks model weights toward zero
C) It increases the learning rate
D) It increases model variance

Correct Answer: B

Explanation: L2 regularization penalizes large weights, reducing model complexity and overfitting.

Question: 13

Given

$$\nabla J = 4,\ \alpha = 2,\ w = 5$$

What is the new gradient after adding L2 regularization?

Options:

A) 6
B) 9
C) 14
D) 15

Correct Answer: C

Question: 14

Using the updated gradient from Q3 and learning rate η=0.1, what is the updated weight value after one step?

Options:

A) 4.6
B) 4.1
C) 3.6
D) 3.0

Correct Answer: C

Question: 15

Which of the following are true effects of applying L2 regularization in this case?

Options:

A) Reduction in overfitting
B) Slight increase in training error
C) Increase in model variance
D) Improvement in test performance

Correct Answers: A, B, D

Explanation: L2 regularization trades a small increase in bias for a significant reduction in variance, improving generalization.